Modal agents in the game of Chicken

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1 Introduction

Setting

Two programs play a one-shot game and each can read the other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent, written with the provability box $\Box$ of the logic GL. Earlier findings on this site study the prisoner's dilemma; here the game is mostly **Chicken**: both swerve 3/3, swerve against straight 1/4, both go straight (crash) 0/0. p means "the opponent swerves against me", q "the opponent swerves against NerveBot", the bot that never swerves. SwerveBot always swerves; FairBot is $\Box p$, graded FairBots are $\Box^j p$. The census covers every agent built from p, q, $\bot$, $\neg$, $\wedge$, $\vee$ and $\Box$ (atoms under a box) up to size 10, 7561 behaviourally distinct agents. Mennen [22] set up modal agents for general games; this finding works out Chicken concretely.

Result

- **Equilibria never crash.** Among the census agents there are 517 symmetric program equilibria, all with mutual swerving, and no equilibrium at all involves a crash. Every asymmetric equilibrium pairs one of the 701 agents that never swerve against anyone in the census with an agent that swerves against it.
- **The smallest sensible Chicken agents** (swerve against themselves and against NerveBot, go straight against SwerveBot) have size 3: exactly $\neg\Box p$ and $\neg\Box q$. The first never crashes against any census agent; the second crashes against 755 of them.
- **Safe and fair.** Exactly 4 census agents never crash against a census agent, go straight against SwerveBot and swerve together with FairBot, and all 4 never crash against any modal opponent at all. The smallest, of size 9, is the $k = 2$ member of a family $Q_k = \Box^k\bot \vee \neg\Box(p \wedge q)$ with exact laws for every k: Q_k never crashes, beats SwerveBot, swerves against itself, swerves together with $\Box^j p$ exactly when $j < k$ and with $\Box(\Box^j\bot \vee p)$ exactly when $j + 2 \leq k$, and goes straight against Q_m for $m > k$.

- **A limit for every two-action game.** If an agent of modal depth d exploits the always-cooperating bot, then graded FairBots $\Box^j p$ with $j \geq d$ end up playing action 2 against it, and the two never settle on mutual action 1. The Q_k family shows the bound is tight.
- **Tournaments and evolution.** In a round robin the top Chicken agent is $\neg\Box p \wedge \Box(\neg p \vee \Box p)$, which crashes against itself. Under replicator dynamics the surviving agents are the same 1116 classes in the prisoner's dilemma, Chicken and Stag Hunt, and they form a maximal set of mutual cooperators containing FairBot.

Why it matters

Modal agents have been studied almost only in the prisoner's dilemma. Chicken punishes mutual defection (a crash) instead of rewarding it, so the question changes from "can I avoid being exploited?" to "can I avoid crashing without being a pushover?". The census answers it concretely: equilibria never crash, safe agents exist and form a family with exact thresholds, and a general depth bound limits how such agents can coexist with graded FairBots.

Proved vs checked

The laws for Q_k and the depth bound are proved for all indices in Lean 4, in the height semantics of closed GL fixed points (see [12]; the link to provability is Solovay's theorem [29]). The census results (equilibria, smallest agents, the four safe agents, round robin, replicator dynamics) hold for all agents up to size 10, computed exactly and re-checked on samples; the replicator results are numerical (2000 steps from the uniform start).

Novelty statement

We believe this result is new. Our prior-art search (2026-10-20) covered:

- Mennen, "Modal combat for games other than the prisoner's dilemma" (LessWrong / Alignment Forum, 2017): framework for modal agents in games with any number of players and actions (player types, ranks, opponent tests naming a target player, an agent to test against and an action); existence and Pareto theorems; no agent families with index laws, no enumeration.
- Gram_Stone, "Modal Chicken" (LessWrong, 2015) and its comments (Alex-Mennen, SilentCal): SwerveBot, NerveBot, CarefulBot, AstuteBot; informal claims about graded CarefulBots and BoldBots.
- Hobson, "Yet More Modal Combat" (LessWrong / Alignment Forum, 2021): a two-system PD bot; generalisation to Chicken and more players left open.

- Critch, Dennis & Russell, "Cooperative and uncooperative institution designs: surprises and problems in open-source game theory" (arXiv 2208.07006): two-player bounded proof-search agents.
- Todorov, ten Napel & Müller, "Parametric Open Source Games" (arXiv 2606.27068): non-modal parametric agents; Chicken and Stag Hunt thresholds.
- Barasz et al. 2014 (arXiv 1401.5577); LaVictoire et al. 2014; Critch, parametric bounded Löb (arXiv 1602.04184); Oesterheld 2022 (arXiv 2211.05057); LessWrong / Alignment Forum wiki "Modal combat"; web searches for modal or provability-based agents in Chicken, Stag Hunt and multiplayer program equilibrium.

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was partly checked by scripts that run Lean proofs (claims K7, K8, K10) and partly checked by script (claims K1, K2, K3, K4); the proof and check files are included so anyone can re-run them.

2 Claims

Claim K1

`In L up to size 10 (7561 classes; Chicken 3/3, 1/4, 0/0), census program equilibria: 517 symmetric, all mutual swerve; none with a crash; every swerve/straight one pairs one of the 701 agents that never swerve against any census agent with an agent swerving against it.`

Status: derived ů author check only.

Verified.

- `checks/K1-c046_games10.py`: (1) class counts; (2) independent re-solve + 13 GLS certificates; (3) Chicken equilibrium counts 517 symmetric all CC, 262741 CC, 2138719 CD/DC, and CD-equilibrium iff straight player never swerves (701 classes) and the other swerves vs it — pass

Evidence.

- reasoning: SwerveBot guarantees at least 1, so a crash is never a best response; a best-response payoff of 1 against b requires that b never swerves.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K2

In L up to size 10, round robin incl. self: Chicken (3,1,4,0) max 17666 only by $\neg\Box p \wedge \Box(\neg p \vee \Box p)$, which crashes vs itself (SwerveBot 16479, FairBot 14303); Stag Hunt (4,0,3,2) max 23411 only by $\Box(p \vee \Box^3\neg(p \vee q))$, which hunts stag vs HareBot (FairBot 21719).

Status: derived ů author check only.

Verified.

- `checks/K1-c046_games10.py`: (5) unique RR maxima 17666 (Chicken) and 23411 (Stag Hunt) with stated agents of size 10 and stated scores of CB, DB, FB; leader cells vs self/DB/CB/FB GLS-certified in (2) — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K3

In L up to size 10, discrete replicator dynamics from uniform start (2000 steps) leave exactly the same 1116 observational classes (weight $> 10^{-6}$) in PD, Chicken and Stag Hunt; they form a maximal set of mutual cooperators containing FairBot.

Status: derived ů author check only.

Verified.

- `checks/K3-c046_replicator10.py`: 3402 observational classes; in each game survivors >1e-6 are the same 1116 (2158 chain classes), others <1e-12, mean payoff R within 1e-9; survivors pairwise CC (exact) and maximal (exact); FB in S, DB not — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K4

In L up to size 10, the smallest agents that swerve vs themselves and vs NerveBot but go straight vs SwerveBot have size 3: exactly $\neg\Box p$ and $\neg\Box q$; $\neg\Box p$ crashes against no census agent, $\neg\Box q$ against 755.

Status: derived ů author check only.

Verified.

- `checks/K1-c046_games10.py`: (6) property set minimum size 3 = {~[]p, ~[]q}; ~[]p no crash vs any of 7561; ~[]q 755 crashes; no census agent anti-coordinates with all others; ~[]p cells vs self/F/~F/[]p GLS-certified in (2) — pass

Evidence.

- reasoning: Never-crash of ~\[\]p against any opponent: going straight means \[\]p holds in the standard model, so p holds by soundness and the opponent swerves.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K7

For all $k \geq 0$, $Q_k = \Box^k \bot \vee \neg\Box(p \wedge q)$ never crashes vs any opponent, beats SwerveBot, self-swerves; it mutually swerves with $\Box^j p$ ($j \geq 1$) iff $j < k$ and with $\Box(\Box^j \bot \vee p)$ iff $j + 2 \leq k$, else is exploited; vs Q_m, $m > k$, Q_k goes straight. Lean.

Status: derived ů author check only.

Verified.

- `checks/K7-c046_chicken_q.py`: (5) Lean 4 core theorems q_safe, q_sb, q_nb, q_fj, q_gj, q_db, q_qm_lt, q_qq, q_bn for all indices and any solving traces, no sorryAx, negative control (j+1<=k for G_j) rejected; (4) height tables for k,j,m,n<=8 agree; (3) GLS certificates for Q_2 cells — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Q_k is straight at h\>=k only if the opponent swerved (and swerved vs NB) at all g\<h; vs \[\]^j p, Q_k's first straight height k is seen by F_j iff k+j\<=h, and F_j's NB trace fails at j, which frees Q_k iff j\<k.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K8

In L up to size 10 (Chicken), exactly 4 agents crash against no census agent, go straight vs SwerveBot and mutually swerve with FairBot; all 4 never crash vs any modal opponent; the unique smallest (size 9) is $\Box^2 \bot \vee \neg\Box(p \wedge q)$.

Status: derived ů author check only.

Verified.

- `checks/K7-c046_chicken_q.py`: (1) 7561 classes, 808 census-safe, 695 SU, SU subset of safe; (2) safe & straight vs CB & CC with FB: exactly 4, sizes 9,10,10,10, all SU, unique of size<=9 is ([][]F|~[](p&q)); Q_0, Q_1 fail only the FB clause; (3) 9 GLS certificates — pass

Evidence.

- reasoning: If Q_2 goes straight at height h\>=2, box(p&q) held there, so the opponent swerved at all g\<h; hence no tail where both go straight.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K10

For every guarded agent A of modal depth d playing action 2 vs the constant action-1 bot: $\Box^j p$ ($j \geq d$) plays 2 vs A from height $d + j$, and $\Box(\Box^j \bot \vee p)$ ($j \geq d - 1$) from height $j + 2$; never mutual action 1 (any 2-action game; Lean). Both tight via Q_k.

Status: derived ů author check only.

Verified.

- `checks/K10-c046_depth_bound.py`: (1) Lean 4 core theorem depth_bound over an inductive formula type (locality + stabilisation lemmas), no sorryAx, two weakened variants rejected; (2) size<=10 census: 7561 classes, constant vs SwerveBot from height md, 3102 straight vs SwerveBot, 12407 (agent,j>=md) pairs with F_j straight from md+j, 92 tight classes, smallest []([]F|~q); (3) explicit height tables — pass (Lean-proved, no sorryAx)
- `checks/K10-c046_depth_bound_g.py`: (1) Lean 4 core: depth_bound, straight_at, depth_bound_G over inductive Fm, no sorryAx, four weakened variants rejected; (2) size<=10 census: 3102 straight-vs-SB classes, 12407 F_j and 12407 G_j pairs straight from md+j resp. j+2, largest CC index vs G_j = vs F_j minus 1, 178 CC, 92 tight (same sets), smallest []([]F|~q); (3) explicit height tables — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Guarded formulas are local; with constant inputs a depth-d formula is constant from height d. F_j swerves unconditionally below j\>=d, so A sees SwerveBot's inputs below d and goes straight at d; F_j sees that from d+j on.
- reasoning: straight_at: inputs equal SwerveBot's below t\>=d, so A is straight at t by locality and stabilisation. G_j plays 1 below j+1 vs A and vs NB; take t=j+1.

Prior art. Possibly new: no prior art found in the owner's search.

Provenance. Run `20261019-1400`, claims K1, K2, K3, K4, K7, K8, K10, run-folder commit `24c2a64bf8b8cb8be184b4d449ce11068d8270ea`, published 2026-10-20.

3 Discussion

The census stops at size 10, and round-robin winners in particular can change as larger agents are added. The language has a single probe (q, how the opponent treats NerveBot); richer probes would give a larger space. The replicator results describe this population, not modal agents in general. A natural next step is Stag Hunt, where the round-robin leader found here is unusual, and a search for threshold families beyond FairBot-like agents there.

4 How to verify

Each finding has its check scripts in `claims/chicken-modal-agents/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

5 Revisions

- 2026-10-21 — Production note: Lean proofs run inside check scripts are now described as such, separately from harness-built proofs

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] Paul Beame and Søren Riis. More on the relative strength of counting principles. In *Proof Complexity and Feasible Arithmetics*, volume 39 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 13–35. American Mathematical Society, 1998.

[3] Christoph Benzmüller. A (simplified) supreme being necessarily exists, says the computer: Computationally explored variants of gödel's ontological argument. In *Proceedings of KR 2020*, 2020.

[4] Christoph Benzmüller. A comment on modal collapse and ultrafilters in gödel's ontological argument. arXiv:2608.07578, 2026.

[5] Christoph Benzmüller. Gödel's and scott's variants of the ontological argument in lean 4 and tptp thf. arXiv:2609.26806, 2026.

[6] Christoph Benzmüller. Proofs without nominals: Gödel's ontological argument, its shallow embedding, and the open questions of the monatshefte notes. arXiv:2609.36279v4, 2026.

[7] Christoph Benzmüller and David Fuenmayor. Computer-supported analysis of positive properties, ultrafilters and modal collapse in variants of gödel's ontological argument. arXiv:1910.08955, 2019.

[8] Christoph Benzmüller and Lawrence C. Paulson. Quantified multimodal logics in simple type theory. *Logica Universalis*, 7(1), 2013.

[9] Christoph Benzmüller and Dana S. Scott. Notes on gödel's and scott's variants of the ontological argument. *Monatshefte für Mathematik*, 2025.

[10] Christoph Benzmüller, Leon Weber, and Bruno Woltzenlogel Paleo. Computer-assisted analysis of the anderson–hájek ontological controversy. *Logica Universalis*, 2017.

[11] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Cambridge University Press, 2001.

[12] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[13] Andrew Critch. A parametric, resource-bounded generalization of löb's theorem, and a robust cooperation criterion for open-source game theory. *Journal of Symbolic Logic*, 2019.

[14] Andrew Critch, Michael Dennis, and Stuart Russell. Cooperative and uncooperative institution designs: Surprises and problems in open-source game theory. arXiv:2208.07006, 2022.

[15] Kit Fine. Propositional quantifiers in modal logic. *Theoria*, 36(3), 1970.

[16] Melvin Fitting. *Types, Tableaus, and Gödel's God*. Kluwer, 2002.

[17] Murdoch J. Gabbay and Dominic P. Mulligan. Nominal Henkin semantics: simply-typed lambda-calculus models in nominal sets. arXiv:1111.0089, 2011.

[18] Gram_Stone. Modal chicken. LessWrong, 2015.

[19] Donald Hobson. Yet more modal combat. LessWrong / AI Alignment Forum, 2021.

[20] Wesley H. Holliday. A note on algebraic semantics for S5 with propositional quantifiers. *Notre Dame Journal of Formal Logic*, 60(2), 2019.

[21] Patrick LaVictoire, Benja Fallenstein, Eliezer Yudkowsky, Mihaly Barasz, Paul Christiano, and Marcello Herreshoff. Program equilibrium in the prisoner's dilemma via löb's theorem. In *AAAI Workshop on Multiagent Interaction without Prior Coordination*, 2014.

[22] Alex Mennen. Modal combat for games other than the prisoner's dilemma. LessWrong / AI Alignment Forum, 2017.

[23] Mladen Mika and Jakob Nordström. A generalized method for proving polynomial calculus degree lower bounds. In *30th Conference on Computational Complexity (CCC 2015)*, volume 33 of *LIPIcs*, pages 467–487, 2015.

[24] Reinhard Muskens. Intensional models for the theory of types. *Journal of Symbolic Logic*, 72(1), 2007.

[25] Caspar Oesterheld. A note on the compatibility of different robust program equilibria of the prisoner's dilemma, 2022.

[26] Alexander A. Razborov. Proof complexity of pigeonhole principles. In *Developments in Language Theory (DLT 2001)*, volume 2295 of *Lecture Notes in Computer Science*, pages 100–116. Springer, 2002.

[27] Søren Riis. *Independence in Bounded Arithmetic*. PhD thesis, University of Oxford, 1993.

[28] Søren Riis and Meera Sitharam. Non-constant degree lower bounds imply linear degree lower bounds. Technical Report TR97-048, Electronic Colloquium on Computational Complexity, 1997.

[29] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.

[30] Aleksandar Todorov, Jesse ten Napel, and Alexander Müller. Parametric open source games. arXiv:2606.27068, 2026.

[31] transhumanist_atom_understander. Is there only one FairBot? LessWrong, with a comment by Zac Hatfield-Dodds, 2026.